VINUNIVERSITY

COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

COMP4010 Project 2 Proposal

Global Population & Demographic Insights Dashboard

Proposal Writeup (490 words)

Course: Data Visualization (COMP4010)
Group: 9
Group Members:

No.	Full Name	Student ID
1	Nguyen Tien Nhan	V202301006
2	Hoang Anh Minh	V202300996
3	Pham Tuan Hung	V202401414
4	Do quang Thai An	V202401422

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This project aims to develop an interactive global demographic visualization dashboard using data from the Our World in Data – Population and Demography Explorer [1]. The dashboard will help users explore and compare worldwide demographic indicators such as birth rate, death rate, life expectancy, fertility rate, and net migration across countries and regions over time.

The main question our project seeks to answer is: *How do demographic patterns reflect different social, economic, and political conditions across the world, and how have these trends evolved over time?* Through an interactive visual platform, users will be able to explore relationships between demographic indicators and uncover contrasting population narratives, such as aging societies with declining fertility rates and conflict-affected regions experiencing migration and mortality disruptions.

This topic matters because demographic changes strongly influence economic development, healthcare systems, labor markets, urbanization, and government policy. Understanding these trends is important not only for researchers and policymakers, but also for students and the general public who want to better understand global population dynamics.

One major challenge of this dataset is its multidimensional nature. The data contains many countries, multiple demographic indicators, and long time ranges spanning decades. Comparing several variables simultaneously across geographic regions can easily become overwhelming in static charts. Additionally, demographic patterns are highly uneven across the world, making it difficult to communicate both global trends and country-level details within a single visualization.

To address these challenges, we propose a multi-level interactive dashboard centered around a 3D rotatable globe interface. When users first enter the website, they will see an interactive globe displaying global demographic data through color gradients and visual markers. Users can rotate and zoom into continents or countries. Clicking on a region will transition the interface into a more detailed regional view, where additional visualizations such as bar charts, line charts, and choropleth maps will appear alongside the selected area.

Users can use the visualization to focus on regions or countries of their interest and derive their own narratives from the data. In addition to supporting open exploration, our project will also highlight several specific demographic stories that emerge clearly from the dataset. One focus group will consist of countries experiencing rapid population aging and low fertility rates, such as Japan, South Korea, and several European nations. Another focus group will examine war-torn countries and regions affected by conflict, where demographic indicators such as death rate, migration, and life expectancy may show significant disruptions over time.

To support these stories, the dashboard will provide optional filters and highlighted views that emphasize selected groups of countries directly on the globe and regional maps. Users may switch between highlighted demographic themes or continue exploring freely through country selection, time sliders, and indicator filters.

The project combines geographic visualization, temporal analysis, and interactive storytelling to create an engaging and informative experience. By transforming complex demographic datasets into intuitive visual interactions, the dashboard aims to make global population trends easier to understand, compare, and interpret.

References

[1] Our World in Data, *Population and Demography Explorer*. Available at: https://ourworldindata.org/explorers/population-and-demography Accessed: May 17, 2026.